Name: Tanvi Shah Roll No: 21CS104

Mid-Term answer sheet

Q1. Overfitting happens when a model learns the training examples too well, including random noise, so it shows very low training error but high error on validation/test data.

1) L1/L2 regularization - adds a penalty for large weights so the model becomes simpler.

2) Data augmentation - creating more varied training examples (flips, crops) so the model can't just memorise them.

Q2. Each token gets Q, K and V vectors by multiplying its embedding with weight matrices that are learned in training. Score = Q.K for every pair of tokens, then softmax converts the scores into weights that sum to one, and the output is the weighted sum of the V vectors. We scale by sqrt(dk) to keep the numbers small.

Q3. P = 40/(40+10) = 0.8

R = 40/(40+20) = 0.67

F1 = 2*0.8*0.67/(0.8+0.67) = 0.73

Accuracy = (40+30)/100 = 0.7

(continued)

Q4. GAN: a generator and a discriminator play a minimax game. Diffusion: the model learns to denoise images that were corrupted with Gaussian noise over many timesteps.
GAN training is unstable and can have mode collapse; diffusion training is stable and the samples are more diverse.
GANs are fast (a single forward pass), diffusion is slow (hundreds of denoising steps).

Q5. RAG: chunk the documents, embed each chunk with an embedding model and store the vectors in a vector DB like FAISS. At query time embed the question, retrieve the top-k similar chunks and put them in the prompt for the LLM.
Failure: chunks too large or too small, so the right information is split or lost -> tune the chunk size and use overlap between chunks.
Another issue is cost.

Name: Farhan Ali Roll No: 21CS105

Mid-Term answer sheet

Q1. Overfitting: the model performs well on training data but fails to generalise to unseen data. We can use dropout and cross validation.

Q2. Self-attention computes attention between all words. Formula: Attention(Q,K,V) = softmax(QK^T/sqrt(dk))V where Q, K, V come from the input through linear layers. The softmax gives the weights. Scaling stops the softmax from saturating and having vanishing gradients when dk is large.

Q3. Precision = 40/50 = 0.8, Recall = 40/60 = 0.67, F1 = (0.8+0.67)/2 = 0.735, Accuracy = 70%

(continued)

Q4. GANs: two models trained against each other. Diffusion models: a forward noising process and a reverse denoising process. GANs are faster at generation. Diffusion models are more stable to train and produce higher diversity.

Q5. RAG = retrieval + generation. Steps: user query -> search the knowledge base -> retrieved text + query given to the LLM -> answer.
Failure 1: wrong documents retrieved -> use re-ranking.
Failure 2: outdated knowledge base -> regularly update the index.

Name: Pooja Nair Roll No: 21CS106

Mid-Term answer sheet

Q1. Overfitting - training accuracy high, test accuracy low; the model learnt the noise.
Methods: regularisation (penalise big weights), getting more training data.

Q2. $Q = XW_q$, $K = XW_k$, $V = XW_v$.
Attention = softmax(QK^T / sqrt(dk)) V.
Softmax makes the weights between 0 and 1.

Q3. P = 0.8, R = 0.667, F1 = 2(0.8)(0.667)/(1.467) = 0.727, acc = 0.7

(continued)

Q4. (a) GAN adversarial, diffusion denoising. (b) GANs unstable. (c) GAN mode collapse. (d) GAN faster.

Q5. Index the docs as embeddings; retrieve the similar chunks for a query; the LLM generates the answer from them.
Failure: hallucination - prompt it to only use the context.